Cyclistic Bike-Share, Case Study 1

Introductions Are in Order

My name is Juliana. I am 36 and currently work as an English teacher in Malta. I am (somewhat humbly) publishing this case study here as a requirement to complete my Google Data Analytics course, as well as to build a portfolio for myself and make a switch — as you might have guessed from the previous reason — into a career in data analytics.

Humbly is the key word here — even if it is in parentheses — and that's because this course was the first time I ever had a go at SQL, Tableau, and all that jazz.

Yes, I used ChatGPT a lot. I still made loads of mistakes, and the road to the conclusions I reached and present here was rocky. But I still believe I found interesting angles, steered the process towards the right questions, and came up with a successful analysis that — if not 100% bulletproof or professionally polished — is still a pretty good one for a newbie.

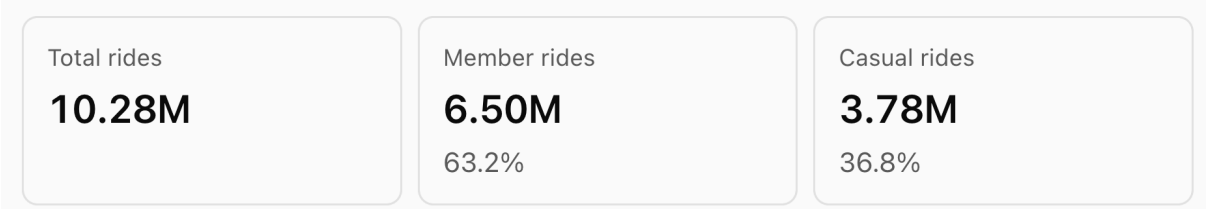
Hope you're able to read it all till the end and enjoy!

Understanding the Differences Between Casual Riders and Annual Members

Analysis period: March 2025 – September 2026

This report examines 10.28 million bike-share trips to understand how annual members and casual riders use Cyclistic differently, and how those insights can inform a marketing strategy to encourage casual riders to purchase annual memberships.

Executive summary



The analysis indicates that annual members use the bike-share system more frequently during weekdays and exhibit commuting-related riding patterns. Casual riders, meanwhile, take longer trips, ride proportionally more on weekends, and more frequently return bicycles to their starting stations.

These findings suggest an opportunity to develop marketing campaigns that emphasize the convenience and potential value of annual membership for casual riders who already use the system for recreational or occasional transportation.

1. Business task

About the company

Cyclistic launched its bike-share program in Chicago in 2016. According to the company background provided for this case study, the program expanded to a fleet of 5,824 bicycles across 692 stations.

The bicycles are geotracked and can be unlocked at one station and returned to another station within the network. This flexibility allows customers to use the service for commuting, recreational cycling, sightseeing, and other transportation needs.

Cyclistic serves two primary customer categories: casual riders, who purchase single-ride or day passes, and annual members, who purchase yearly memberships.

Business problem

Cyclistic's marketing strategy aims to increase the number of annual memberships. Rather than focusing exclusively on attracting new customers, the company wants to understand how existing casual riders might be encouraged to become members.

The central business question is:

> *How do annual members and casual riders use Cyclistic bikes differently, and how can these differences inform strategies to convert casual riders into annual members?*

The objectives are to identify differences in riding frequency, trip duration, seasonal behavior, weekday and weekend usage, popular stations, and route preferences, then translate those findings into actionable marketing recommendations.

2. Data sources

The analysis uses historical bike-share trip records from March 2025 through September 2026, covering 19 months. This period comprises 19 months of the most recent data concerning the trips log

The monthly CSV files were imported into Google BigQuery and combined into a single analytical dataset.

The principal fields used were:

Field	Purpose
ride_id	Identify individual trips
started_at	Determine trip start time, date and weekday
ended_at	Calculate trip duration
member_casual	Distinguish members from casual riders
start_station_name	Identify popular trip origins
end_station_name	Identify popular trip destinations
start_station_id	Identify starting stations
end_station_id	Identify ending stations

The dataset contains trip-level records rather than customer profiles. It therefore cannot directly establish individual rider motivations, income, demographics, or whether a casual rider subsequently purchases a membership.

3. Data cleaning and manipulation

Google BigQuery was used to prepare and analyze the dataset, with Google Sheets supporting the review and visualization of query results.

The preparation workflow included combining monthly records, checking data types, investigating missing station values, and identifying invalid trip durations.

The cleaning operations discussed during the analysis included:

- Removing records with missing start or end timestamps.
- Excluding trips with zero or negative durations.
- Checking for possible duplicate trip records.
- Accounting for missing station names when analyzing station popularity.
- Combining the monthly data into one dataset.
- Creating derived variables for trip duration, season, day of week, weekday/weekend classification, and hour of day.

For example, trip duration was calculated in minutes using the difference between ended_at and started_at.

The dataset was then grouped by rider type, season, weekday, hour, and station to produce summary statistics.

3.1 Tools, AI Assistance, and Analytical Validation

This project was conducted using Google BigQuery for data preparation and SQL-based analysis, with Google Sheets used to organize and review results.

ChatGPT (OpenAI) was used as an AI-assisted analytical tool throughout the project, primarily to:

- Support the development and troubleshooting of SQL queries in BigQuery.
- Explain SQL functions and analytical calculations.
- Suggest appropriate visualization techniques for different types of data.
- Assist with the interpretation and presentation of analytical findings.
- Support the organization and drafting of the final report.

The analytical questions, dataset selection, execution of queries, and review of results were carried out by the analyst. AI-generated suggestions were used to support the analytical workflow rather than as independent evidence.

The conclusions presented in this report are based on the dataset and query outputs, not on AI-generated estimates or illustrative visualizations.

AI assistance also introduced a need for careful verification. Query logic, calculations, visualizations, and interpretations were thoroughly reviewed against the underlying data before publication.

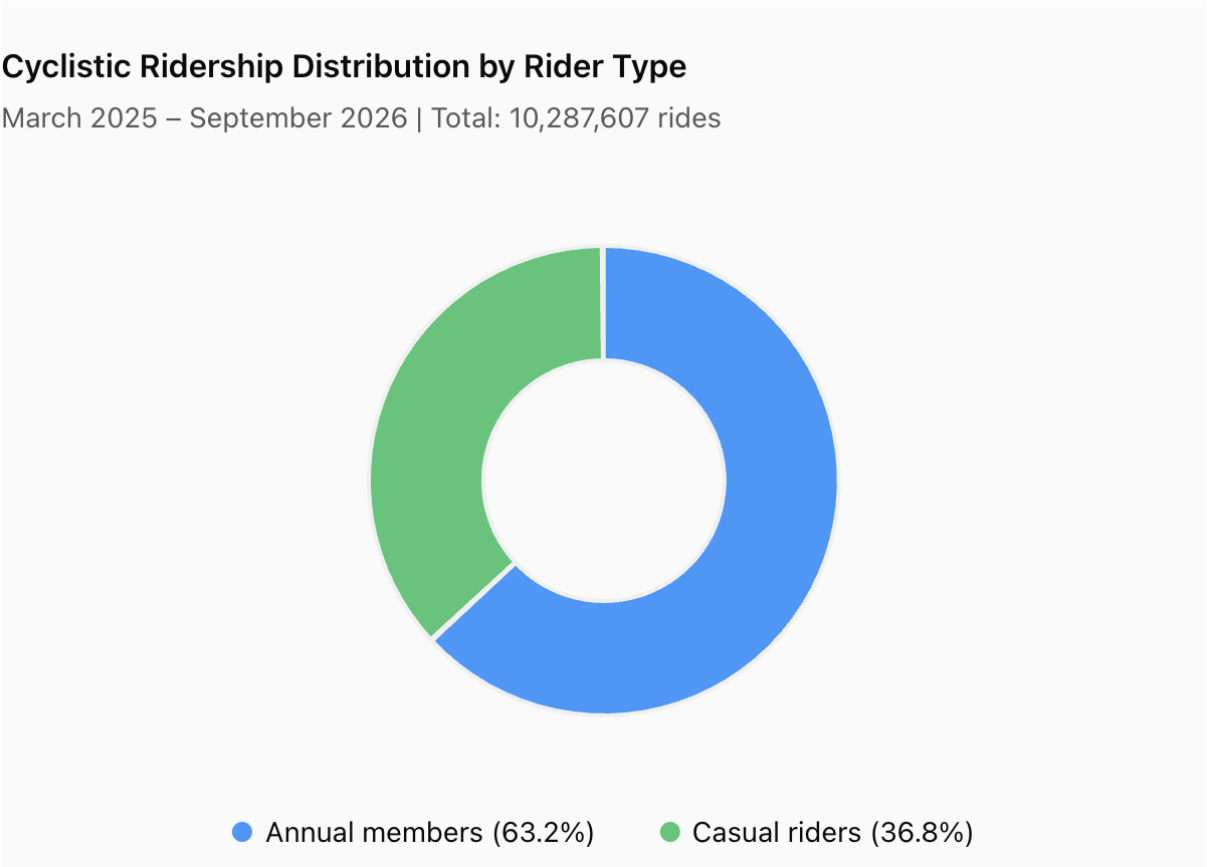
4. Summary of analysis

4.1 Membership distribution

Distribution of Trips by Rider Type

March 2025–September 2026.

Across the combined period, annual members accounted for 6,503,708 trips (63.2%), while casual riders accounted for 3,783,899 trips (36.8%).

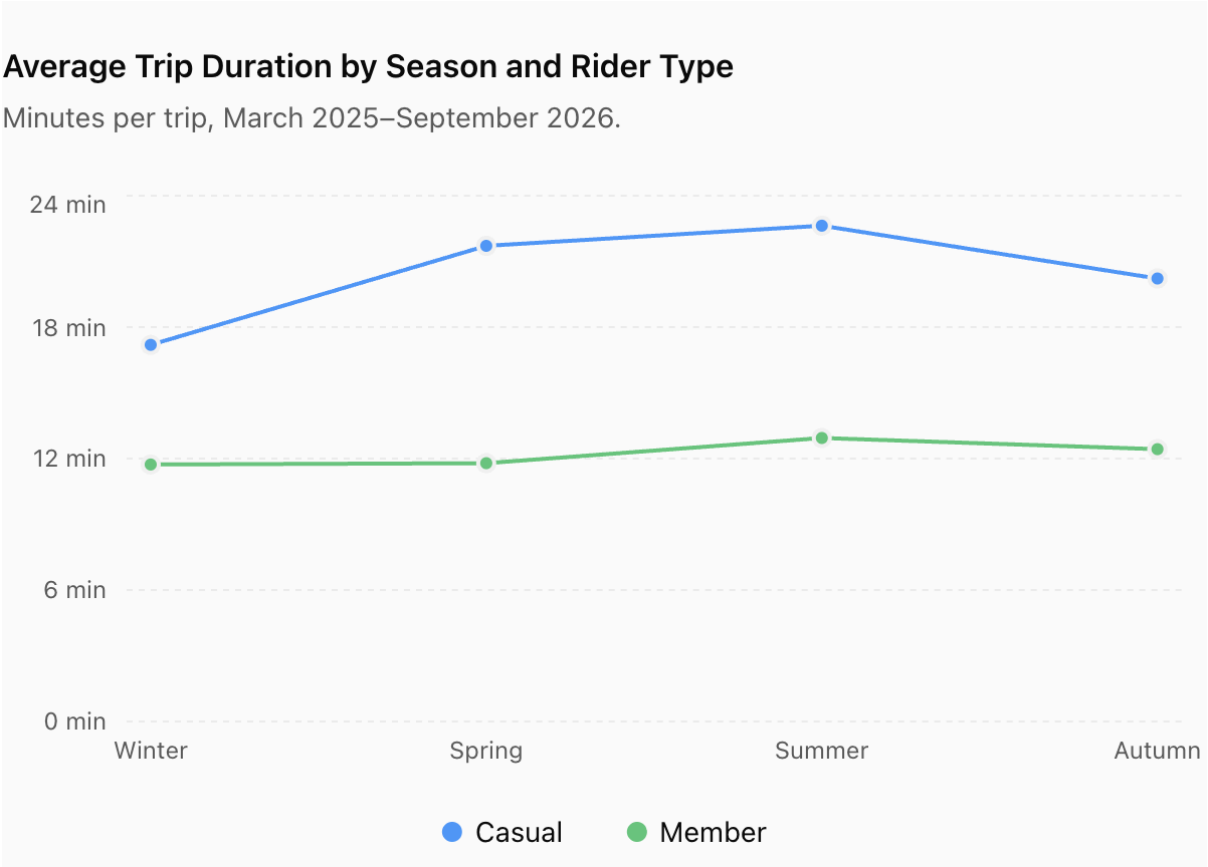


This shows that members generate the majority of recorded trips. However, trip counts do not equal the number of unique customers, so these figures cannot establish how many individuals belong to each group.

4.2 Trip duration

Average Trip Duration by Season

Minutes per ride, comparing annual members and casual riders.



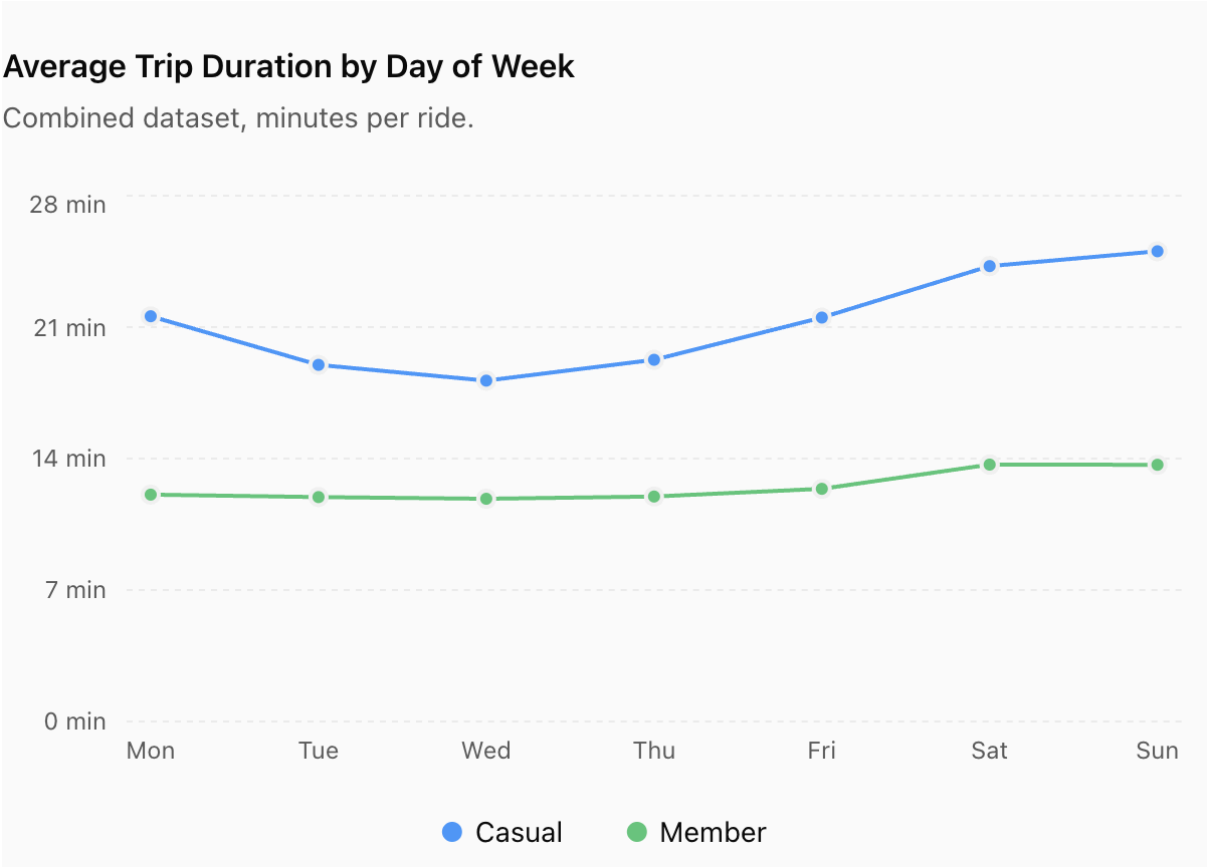
Casual riders consistently take longer trips than members across all four seasons. In summer, casual trips average 22.63 minutes, compared with 12.94 minutes for members, making them approximately 75% longer.

Member trip durations remain comparatively stable throughout the year. Casual trip durations increase during spring and summer, suggesting that warmer weather may encourage longer journeys.

4.3 Weekday and weekend behavior

Average Trip Duration: Weekdays vs. Weekends

Minutes per ride.



Weekend rides account for approximately 37.5% of casual trips but only 23.6% of member trips.

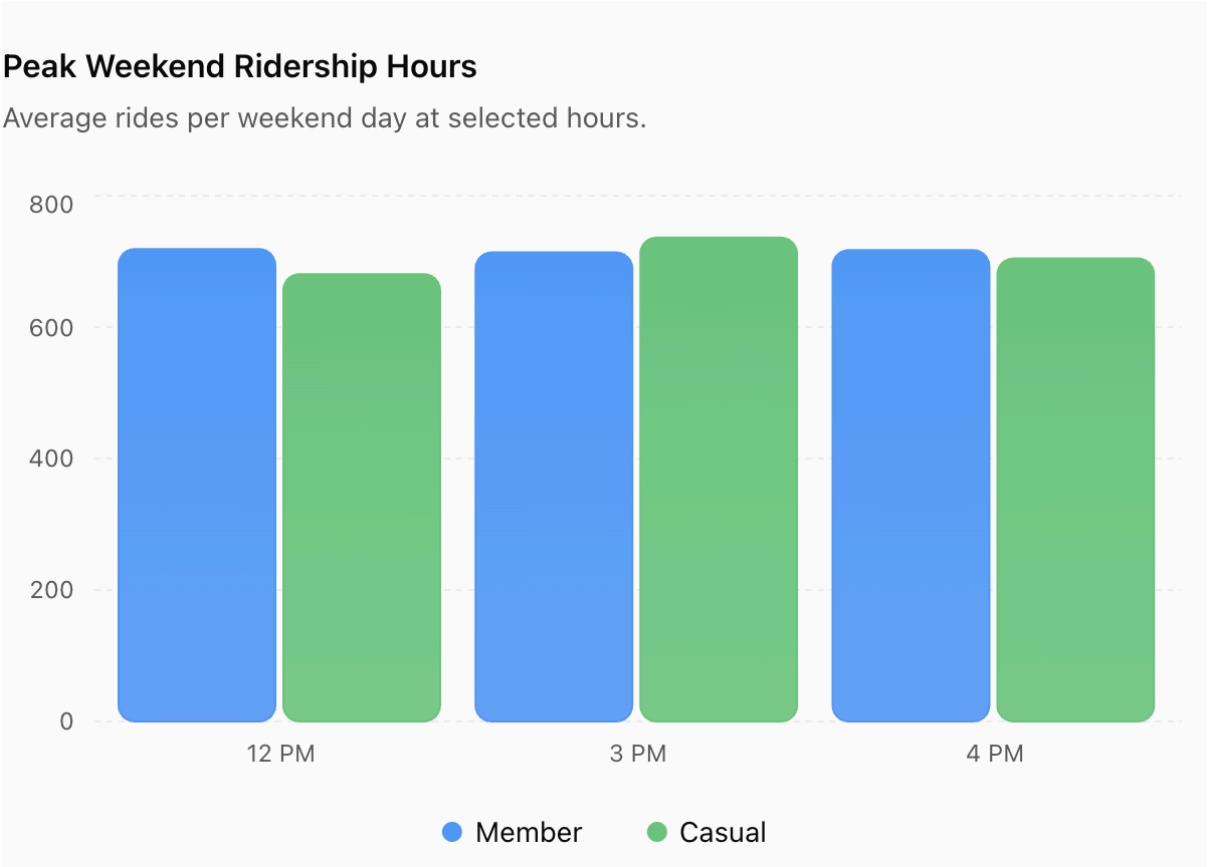
Casual riders also increase their average trip duration from 20.01 minutes on weekdays to 24.60 minutes on weekends, an increase of approximately 23%.

These results suggest that casual riders are more strongly associated with weekend activity, whereas members use the service more consistently throughout the working week.

4.4 Hourly ridership patterns

The hourly analysis provides additional evidence of differences in usage patterns.

Members demonstrate distinct weekday peaks around 8 AM and 5 PM, which are consistent with typical commuting hours. Casual riders show stronger afternoon activity, particularly on weekends.



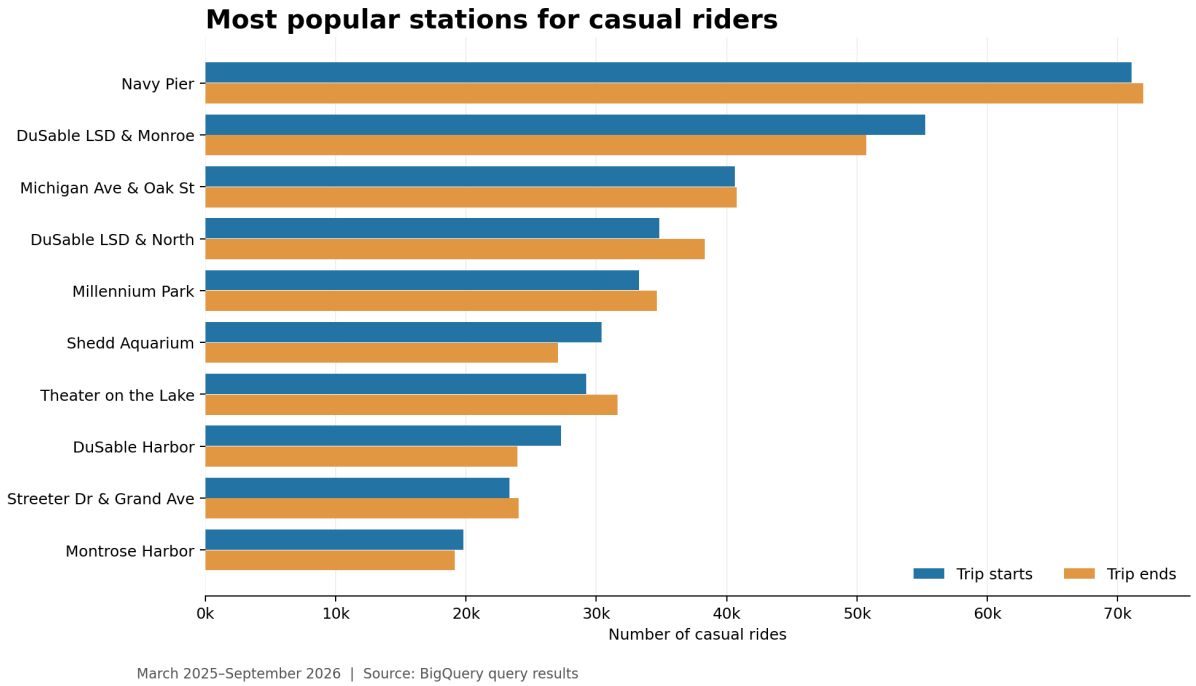
The average-rides-per-day analysis is more appropriate for comparing weekdays and weekends because it accounts for the different number of days in each category.

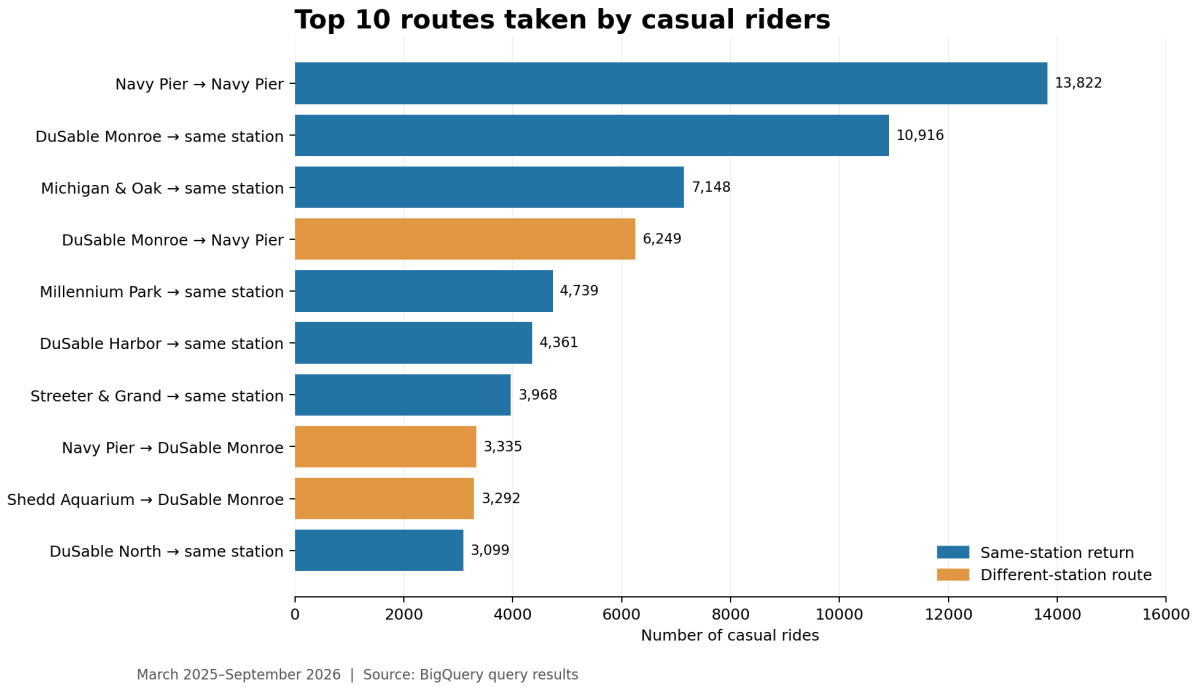
Together, these patterns suggest that routine transportation may play a greater role in member ridership, while recreational activities may be more prominent among casual riders.

4.5 Station popularity and route preferences

Top 10 Starting Stations for Casual Riders

Combined March 2025–September 2026 data. Names shortened for readability.

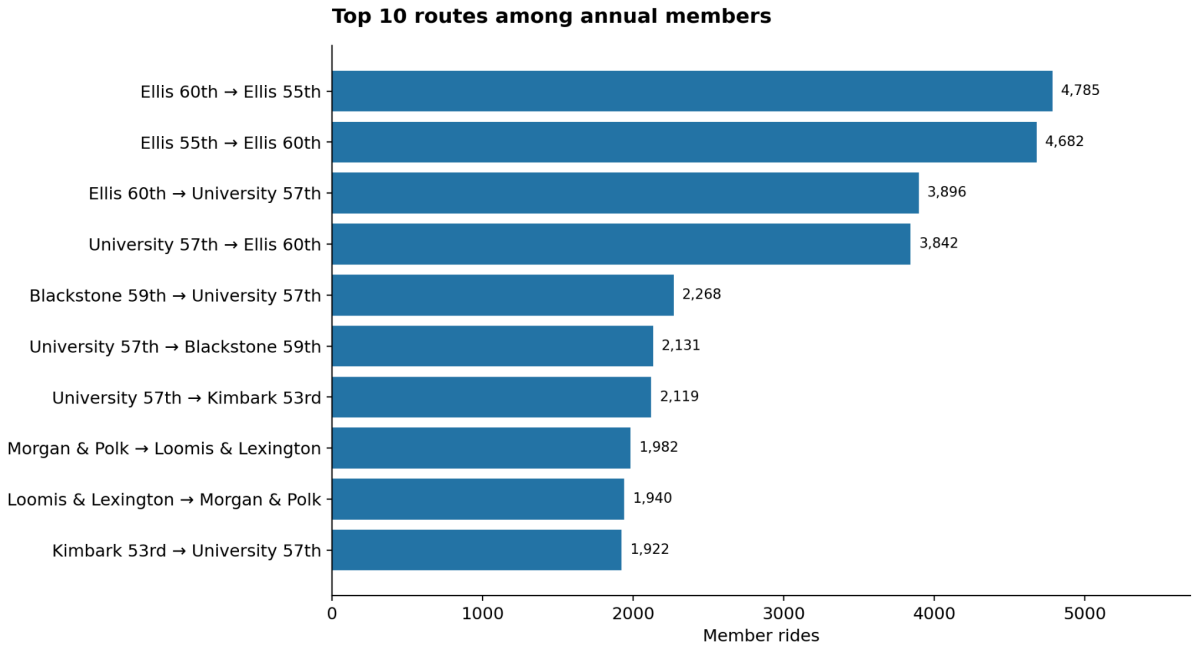


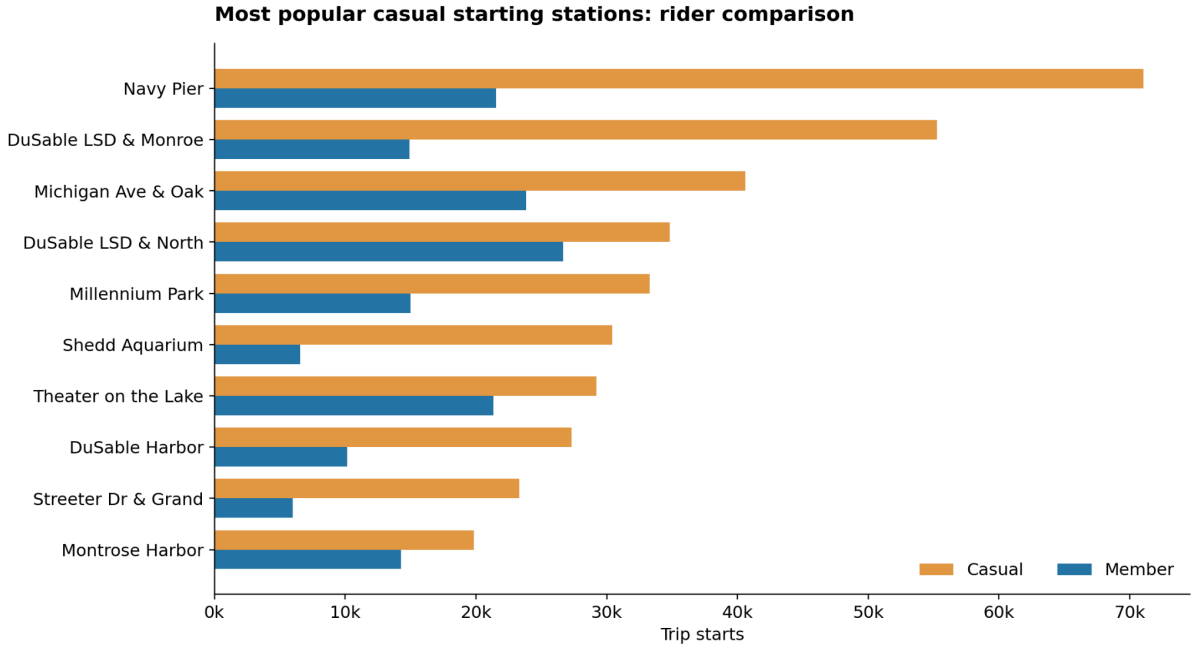


The most popular casual starting stations are concentrated around Chicago's lakefront and major visitor attractions.

Navy Pier ranks first, with 71,051 casual trip starts, followed by DuSable Lake Shore Drive & Monroe Street with 55,247.

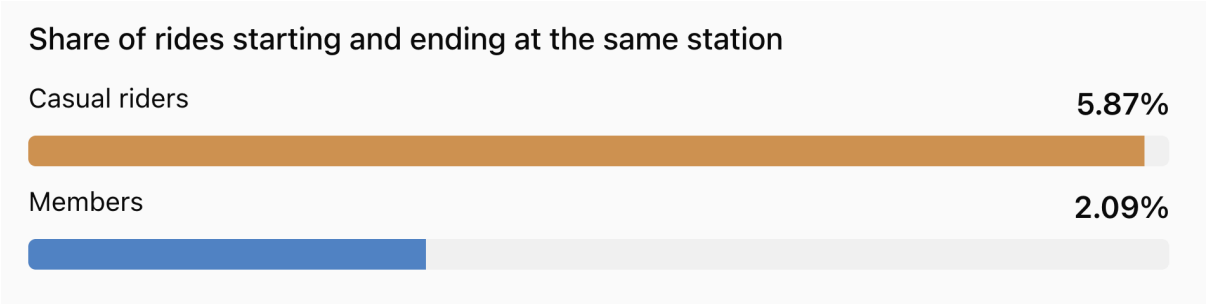
In contrast, many of the most popular member routes connect stations in the Hyde Park area, including Ellis Avenue and University Avenue.





These geographic patterns are consistent with different travel purposes, although the data does not directly identify why a trip occurred.

4.6 Same-station returns



Rider type	Same-station rides	Percentage
Casual	222,015	5.87%
Member	136,062	2.09%

Among casual rides, 222,015 trips (5.87%) returned to the starting station. Among member rides, 136,062 trips (2.09%) did so.

Casual rides were approximately 2.8 times as likely to end at their starting station.

This is consistent with the hypothesis that recreational or circular journeys play a larger role in casual ridership.

5. Key findings

The analysis identifies five main findings:

1. Annual members account for most recorded trips, representing 63.2% of ridership.
2. Casual riders take longer trips across every season and throughout the week.
3. Casual ridership is more weekend-oriented, with longer average durations on Saturdays and Sundays.
4. Members demonstrate commuting-related hourly patterns, particularly on weekdays.
5. Casual riders favor lakefront stations and same-station returns, consistent with greater recreational usage.

Taken together, these findings suggest that Cyclistic's two customer categories exhibit meaningfully different riding patterns.

6. Top three recommendations

1. Target casual riders at popular recreational destinations

Cyclistic should test location-based membership promotions at high-traffic casual stations, particularly Navy Pier, Millennium Park, and lakefront destinations.

Marketing materials could explain how annual membership benefits frequent recreational cyclists, with QR codes linking to membership information.

Success metric: Membership conversions attributable to station-specific campaigns.

2. Launch seasonal and weekend-focused campaigns

Casual riding activity is particularly concentrated during warmer months and weekends. Cyclistic should prioritize membership acquisition campaigns before and during spring and summer.

Campaigns could highlight the convenience of spontaneous rides and potential savings for customers who ride frequently. Any savings claims should be supported by actual membership and pass prices.

Success metric: Membership conversion rate and acquisition cost during targeted campaign periods.

3. Promote membership as an everyday transportation option

Members show stronger weekday and commuting-hour patterns. Cyclistic could test messaging that encourages casual riders to consider using bikes for routine transportation in addition to leisure activities.

Campaigns could highlight convenience, transportation flexibility, and the benefits of regular cycling. Testing different messages would help determine which appeals are most effective.

Success metric: Membership conversion rates from commuting-focused campaigns, compared with a control group.

7. Conclusion

The analysis of approximately 10.28 million Cyclistic trips from March 2025 through September 2026 reveals consistent differences between casual riders and annual members.

Members account for the majority of recorded trips and demonstrate stronger weekday and commuting-hour usage patterns. Casual riders take longer journeys, are more active on weekends, and are more likely to start and finish at the same station.

These differences suggest that Cyclistic could benefit from a more targeted marketing strategy that recognizes casual riders' existing preferences while communicating the potential benefits of annual membership.

By focusing on recreational destinations, seasonal campaigns, and everyday transportation opportunities, Cyclistic can test strategies intended to encourage more casual riders to become annual members.

Ultimately, the effectiveness of these recommendations should be evaluated through measurable conversion rates, campaign experiments, and customer feedback rather than inferred from trip behavior alone.

8. Data Manipulation Log (SQL Queries)

1. Combining all months into one table

```
CREATE OR REPLACE TABLE `deft-condition-312117.casestudy1.202503_202609` AS

SELECT * FROM `deft-condition-312117.casestudy1.2025FULL`
UNION ALL
SELECT * FROM `deft-condition-312117.casestudy1.2026_full`
```

2. Cleaning

Checking how many rows contain NULL start or ending times:

```
SELECT
 COUNT(*) AS trips_with_null_timestamps
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE started_at IS NULL
  OR ended_at IS NULL;
```

Delete invalid trips (trips with under a minute duration):

```
DELETE FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE started_at IS NULL
  OR ended_at IS NULL
  OR TIMESTAMP_DIFF(ended_at, started_at, SECOND) <= 0;
```

Checking for duplicate rows: (zero occurrences)

```
SELECT
 TO_JSON_STRING(t) AS row_data,
 COUNT(*) AS occurrences
FROM `deft-condition-312117.casestudy1.202503_202609` AS t
GROUP BY row_data
HAVING COUNT(*) > 1
ORDER BY occurrences DESC;
```

Checking for duplicate trip IDs: (zero occurrences)

```
CREATE OR REPLACE TABLE `deft-condition-312117.casestudy1.202503_202609` AS
SELECT *
FROM `deft-condition-312117.casestudy1.202503_202609`
QUALIFY ROW_NUMBER() OVER (
 PARTITION BY ride_id
 ORDER BY started_at
) = 1;
```

3. Analysis queries:

Total trips (member and casual) per season:

```
SELECT
 CASE
   WHEN EXTRACT(MONTH FROM started_at)
     IN (12, 1, 2) THEN 'Winter'
   WHEN EXTRACT(MONTH FROM started_at)
     IN (3, 4, 5) THEN 'Spring'
   WHEN EXTRACT(MONTH FROM started_at)
     IN (6, 7, 8) THEN 'Summer'
   ELSE 'Autumn'
 END AS season,
 member_casual,
 COUNT(*) AS total_rides
FROM `deft-condition-312117.casestudy1.202503_202609`
GROUP BY season, member_casual
ORDER BY season, member_casual;
```

Checking for average trip time per season:

```
SELECT
 CASE
   WHEN EXTRACT(MONTH FROM started_at) IN (12, 1, 2)
     THEN 'Winter'
   WHEN EXTRACT(MONTH FROM started_at) IN (3, 4, 5)
     THEN 'Spring'
   WHEN EXTRACT(MONTH FROM started_at) IN (6, 7, 8)
     THEN 'Summer'
```

```
    ELSE 'Autumn'
  END AS season,

  ROUND(
    AVG(TIMESTAMP_DIFF(ended_at, started_at, SECOND)) / 60,
    2
  ) AS avg_trip_minutes
  FROM `deft-condition-312117.casestudy1.202503_202609`

WHERE started_at IS NOT NULL
  AND ended_at > started_at

GROUP BY season
ORDER BY
  CASE season
    WHEN 'Winter' THEN 1
    WHEN 'Spring' THEN 2
    WHEN 'Summer' THEN 3
    WHEN 'Autumn' THEN 4
  END;
```

Comparing seasonal average trip time between member and casual trips:

```
SELECT
  CASE
    WHEN EXTRACT(MONTH FROM started_at) IN (12, 1, 2)
      THEN 'Winter'
    WHEN EXTRACT(MONTH FROM started_at) IN (3, 4, 5)
      THEN 'Spring'
    WHEN EXTRACT(MONTH FROM started_at) IN (6, 7, 8)
      THEN 'Summer'
    ELSE 'Autumn'
  END AS season,

  member_casual,

  COUNT(*) AS total_rides,

  ROUND(
    AVG(TIMESTAMP_DIFF(ended_at, started_at, SECOND)) / 60,
```

```
   2
 ) AS avg_trip_minutes
FROM `deft-condition-312117.casestudy1.202503_202609`

WHERE started_at IS NOT NULL
 AND ended_at > started_at

GROUP BY season, member_casual

ORDER BY
 CASE season
   WHEN 'Winter' THEN 1
   WHEN 'Spring' THEN 2
   WHEN 'Summer' THEN 3
   WHEN 'Autumn' THEN 4
 END,
 member_casual;
```

Checking for the most popular start station for casual rides:

```
SELECT
 start_station_name,
 COUNT(*) AS total_rides
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE member_casual = 'casual'
 AND start_station_name IS NOT NULL
GROUP BY start_station_name
ORDER BY total_rides DESC
LIMIT 10;
```

Checking for the most popular end station for casual rides:

```
SELECT
 end_station_name,
 COUNT(*) AS total_rides
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE member_casual = 'casual'
 AND end_station_name IS NOT NULL
GROUP BY end_station_name
ORDER BY total_rides DESC
```

```
LIMIT 10;
```

Checking for same start and end station for casual rides:

```
SELECT
 start_station_name,
 end_station_name,
 COUNT(*) AS total_rides
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE member_casual = 'casual'
 AND start_station_name IS NOT NULL
 AND end_station_name IS NOT NULL
GROUP BY
 start_station_name,
 end_station_name
ORDER BY total_rides DESC
LIMIT 10;
```

Comparing member and casual total trips by start station:

```
SELECT
 start_station_name,
 COUNTIF(member_casual = 'casual') AS casual_rides,
 COUNTIF(member_casual = 'member') AS member_rides,
 COUNT(*) AS total_rides
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE start_station_name IS NOT NULL
 AND member_casual IN ('casual', 'member')
GROUP BY start_station_name
ORDER BY casual_rides DESC
LIMIT 10;
```

Comparing same start-end stations between member and casual rides:

```
SELECT
 member_casual,
 COUNT(*) AS total_rides,

 COUNTIF(
   start_station_id = end_station_id
```

```
    AND start_station_id IS NOT NULL
  ) AS same_station_rides,

  ROUND(
    100 * SAFE_DIVIDE(
      COUNTIF(
        start_station_id = end_station_id
        AND start_station_id IS NOT NULL
      ),
      COUNT(*)
    ),
    2
  ) AS same_station_percentage
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE member_casual IN ('casual', 'member')

GROUP BY member_casual;
```

Checking the most popular start and end stations for members:

```
SELECT
  start_station_name,
  end_station_name,
  COUNT(*) AS total_rides
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE member_casual = 'member'
  AND start_station_name IS NOT NULL
  AND end_station_name IS NOT NULL
GROUP BY
  start_station_name,
  end_station_name
ORDER BY total_rides DESC
LIMIT 10;
```

Comparing total rides and average trip time between member and casual trips per each day of the week

```
SELECT
  FORMAT_DATE('%A', DATE(started_at)) AS day_of_week,
  member_casual,
```

```
 COUNT(*) AS total_rides,

 ROUND(
   AVG(
     TIMESTAMP_DIFF(ended_at, started_at, SECOND)
   ) / 60,
   2
 ) AS avg_trip_minutes
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE started_at IS NOT NULL
 AND ended_at > started_at
 AND member_casual IN ('member', 'casual')

GROUP BY day_of_week, member_casual

ORDER BY
 CASE day_of_week
   WHEN 'Monday' THEN 1
   WHEN 'Tuesday' THEN 2
   WHEN 'Wednesday' THEN 3
   WHEN 'Thursday' THEN 4
   WHEN 'Friday' THEN 5
   WHEN 'Saturday' THEN 6
   WHEN 'Sunday' THEN 7
 END,
 member_casual;
```

Comparing total trips and average time between casual and member trips between weekdays and weekends

```
SELECT
 CASE
   WHEN EXTRACT(DAYOFWEEK FROM started_at)
     IN (1, 7) THEN 'Weekend'
   ELSE 'Weekday'
 END AS day_type,

 member_casual,

 COUNT(*) AS total_rides,
```

```
 ROUND(
   AVG(
     TIMESTAMP_DIFF(ended_at, started_at, SECOND)
   ) / 60,
   2
 ) AS avg_trip_minutes
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE started_at IS NOT NULL
 AND ended_at > started_at

GROUP BY day_type, member_casual
ORDER BY day_type, member_casual;
```

Comparing total trips between casual and member trips per hour of day:

```
SELECT
 EXTRACT(HOUR FROM started_at) AS hour_of_day,
 member_casual,
 COUNT(*) AS total_rides
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE started_at IS NOT NULL
 AND member_casual IN ('member', 'casual')

GROUP BY hour_of_day, member_casual
ORDER BY hour_of_day, member_casual;
```

Comparing total trips between casual and member trips per hour of day, between weekdays and weekend:

```
WITH daily_hourly AS (
 SELECT
   DATE(started_at) AS ride_date,

   CASE
     WHEN EXTRACT(DAYOFWEEK FROM started_at) IN (1, 7)
       THEN 'Weekend'
     ELSE 'Weekday'
   END AS day_type,
```

```sql
    EXTRACT(HOUR FROM started_at) AS hour_of_day,

    member_casual,

    COUNT(*) AS rides
FROM `deft-condition-312117.casestudy1.202503_202609`
WHERE started_at IS NOT NULL
    AND member_casual IN ('member', 'casual')

 GROUP BY 1, 2, 3, 4),

calendar AS (
 SELECT
    ride_date,
    CASE
      WHEN EXTRACT(DAYOFWEEK FROM ride_date) IN (1, 7)
        THEN 'Weekend'
      ELSE 'Weekday'
    END AS day_type
 FROM UNNEST(
    GENERATE_DATE_ARRAY('2025-03-01', '2026-09-30')
 ) AS ride_date
)

SELECT
 c.day_type,
 h.hour_of_day,
 h.member_casual,

 ROUND(
    SUM(COALESCE(d.rides, 0)) /
    COUNT(DISTINCT c.ride_date),
    2
 ) AS avg_rides_per_day

FROM calendar c

CROSS JOIN (
 SELECT hour_of_day, member_casual
```

```sql
 FROM UNNEST(GENERATE_ARRAY(0, 23)) AS hour_of_day
 CROSS JOIN UNNEST(['member', 'casual']) AS member_casual
) h

LEFT JOIN daily_hourly d
 ON c.ride_date = d.ride_date
 AND h.hour_of_day = d.hour_of_day
 AND h.member_casual = d.member_casual

WHERE c.day_type = CASE
 WHEN EXTRACT(DAYOFWEEK FROM c.ride_date) IN (1, 7)
   THEN 'Weekend'
 ELSE 'Weekday'
END

GROUP BY c.day_type, h.hour_of_day, h.member_casual
ORDER BY c.day_type, h.hour_of_day, h.member_casual;
```